

Maven + java

Pipeline script

```
pipeline {
  agent any

  tools {
    maven 'M3'
  }

  stages {

    stage('Checkout') {
      steps {
        git branch: 'main', url: 'your-repo-url'
      }
    }

    stage('Build & Test') {
      steps {
        bat 'mvn clean package'
      }
    }

    stage('Docker Build') {
      steps {
        bat 'docker build -t my-java-app .'
      }
    }

    stage('Docker Run') {
      steps {
        bat 'docker run -d -p 8080:8080 my-java-app'
      }
    }
  }
}
```

docker file

```
FROM openjdk:17
WORKDIR /app
COPY target/*.jar app.jar
CMD ["java", "-jar", "app.jar"]
```

html pipeline script

```
pipeline {
  agent any

  environment {
    IMAGE_NAME = "html-app"
    DOCKER_HUB = "your-dockerhub-username"
  }

  stages {

    stage('Checkout Code') {
      steps {
        git branch: 'main', url: 'YOUR_GITHUB_REPO_URL'
      }
    }

    stage('Build Docker Image') {
      steps {
        bat "docker build -t %DOCKER_HUB%/%IMAGE_NAME%:latest ."
      }
    }

    stage('Push to Docker Hub') {
      steps {
        withCredentials([string(credentialsId: 'dockerhub-password', variable:
'DOCKER_PASS')]) {
          bat "docker login -u %DOCKER_HUB% -p %DOCKER_PASS%"
          bat "docker push %DOCKER_HUB%/%IMAGE_NAME%:latest"
        }
      }
    }

    stage('Deploy to Kubernetes') {
      steps {
        bat "kubectl apply -f deployment.yaml"
      }
    }
  }
}
```

docker file

```
FROM nginx:latest

# Remove default nginx content
RUN rm -rf /usr/share/nginx/html/*

# Copy your website files
COPY . /usr/share/nginx/html
```

```
# Expose port
EXPOSE 80
```

```
CMD ["nginx", "-g", "daemon off;"]
```

deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: html-app-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: html-app
  template:
    metadata:
      labels:
        app: html-app
    spec:
      containers:
        - name: html-container
          image: your-dockerhub-username/html-app:latest
          ports:
            - containerPort: 80
```

service.yaml

```
apiVersion: v1
kind: Service
metadata:
  name: html-app-service
spec:
  type: NodePort
  selector:
    app: html-app
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
      nodePort: 30007
```